

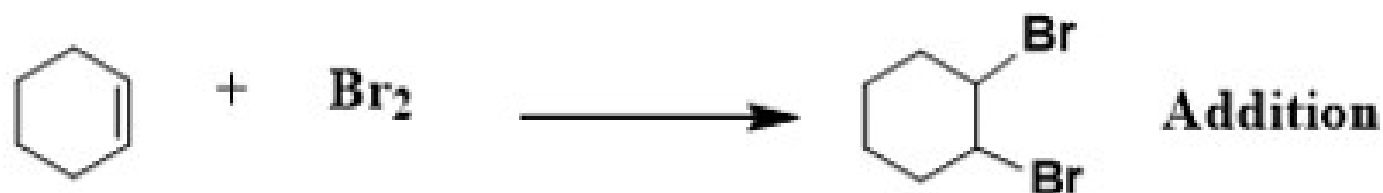
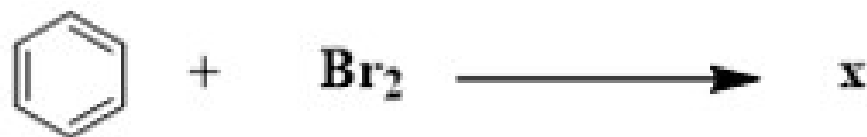
UNIT 5

AROMATIC CHEMISTRY

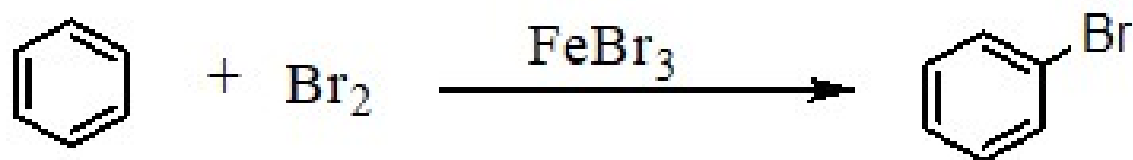
5.1 Aromaticity

Benzene has double bonds, but it shows unusual properties unlike alkenes.

- It has high degree of unsaturation (double bond), but resistant to addition reaction



- It is generally less reactive than alkenes
- It reacts with Br_2 in the presence of Fe catalyst = substitution reaction

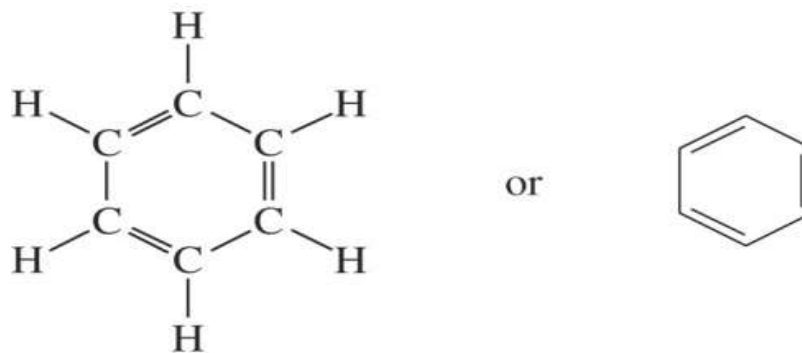


These properties makes benzene different from alkene

5.2 Properties of Benzene and its Derivatives

In 1865 August Kekule proposed the structure of benzene (C_6H_6)

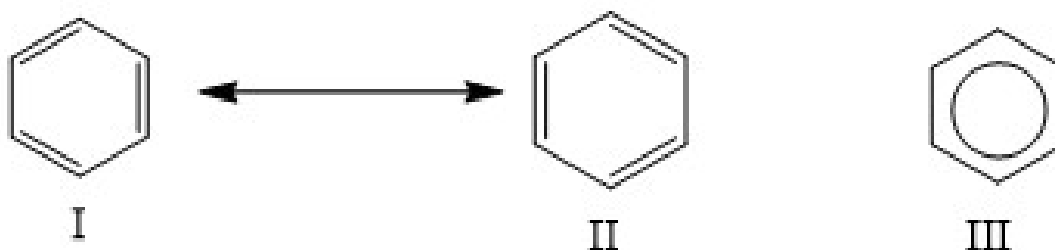
- All six carbon atoms are in a ring
- All c-atoms bonded to each other by alternating single and double bond
- One H-atom is attached to each c-atom
- All H-atoms are equivalent



The Kekulé formula for benzene

Modern Theories of the Structure of Benzene

- Benzene is particularly stable because it has two equivalent and important resonance structures
- Each carbon-carbon bond is $1.39 \text{ \AA}$, which is between the length of a carbon-carbon single bond ($1.47 \text{ \AA}$) and a carbon-carbon double bond ($1.33 \text{ \AA}$)
- Often the hybrid is represented by a circle in a hexagon (III)



Structures I and II are equal resonance contributors to the real structure of benzene

General properties of aromatic compounds

- They have high degree of unsaturation, but resistant to addition reaction
- They favor electrophilic substitution reaction
- They are cyclic compound
- They are flat and planar

Huckel's Rule:

The $4n+2\pi$ Electron Rule

- Planar monocyclic rings with a continuous system of p orbitals and $4n + 2\pi$ electrons are aromatic ($n = 1, 2, 3 \dots$)



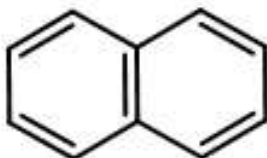
$6\pi \text{ e}^-$

$$\Rightarrow 4(1) + 2 = 6$$

$$\Rightarrow n=1$$

Aromatic

it is planar, resonating e^-



$10\pi \text{ e}^-$

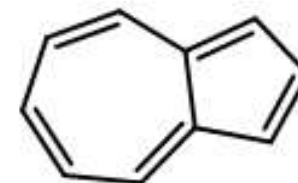
$$\Rightarrow n=2$$

Aromatic



non aromatic

the $\pi \text{ e}^-$ are not resonating



aromatic

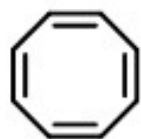
The Annulenes

- are monocyclic compounds with alternating double and single bonds
- Annulenes are named using a number in brackets that indicates the ring size
- An annulene is aromatic if it has $4n+2\pi$ electrons and a planar carbon



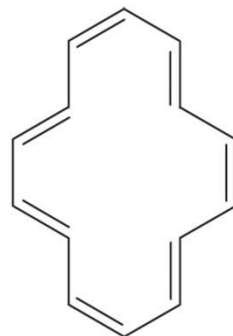
Benzene

[6] Annulene
Aromatic

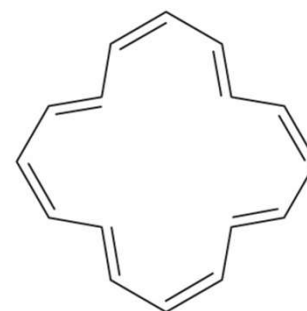


cyclooctatetraene

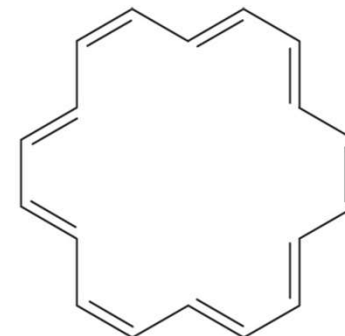
[8] Annulene
Non aromatic



[14]Annulene
(aromatic)



[16]Annulene
(not aromatic)



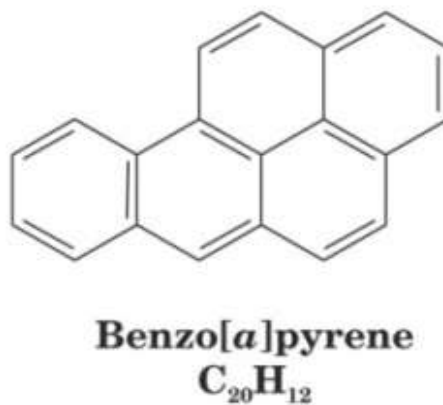
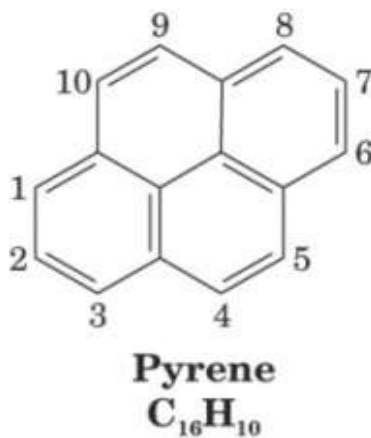
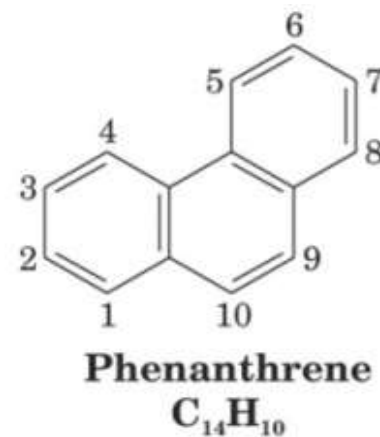
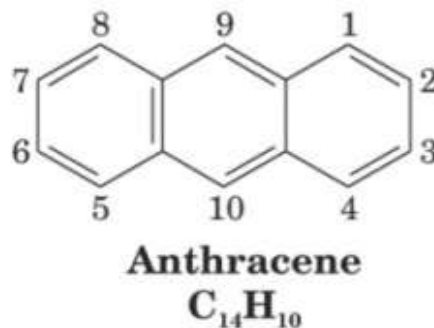
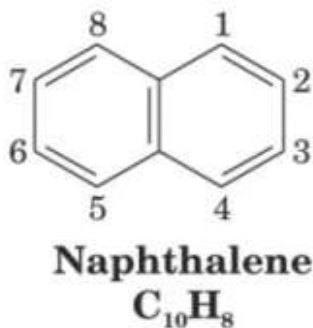
[18]Annulene
(aromatic)

The [14] and [18] annulenes are aromatic ($4n+2$, where $n=3, 4$).

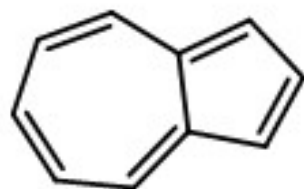
The [16] annulene is not aromatic

Other Aromatic Compounds

- **Benzenoid Aromatic Compounds:** have two or more benzene rings fused together



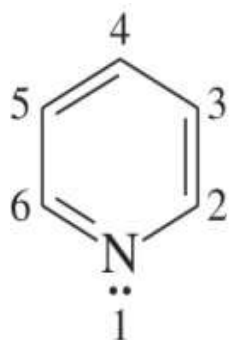
- **Nonbenzenoid Aromatic Compounds:** have two or more rings fused together, but none of the ring is a benzene structure, confirm to Huckel's rule



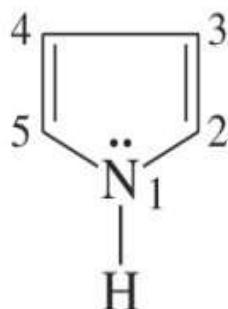
Azulene

Heterocyclic Aromatic Compounds

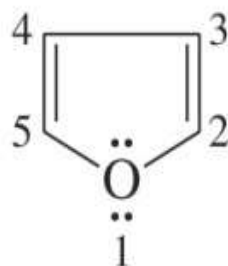
- have an element other than carbon as a member of the ring.
- Numbering always starts at the heteroatom



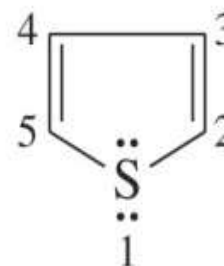
Pyridine



Pyrrole



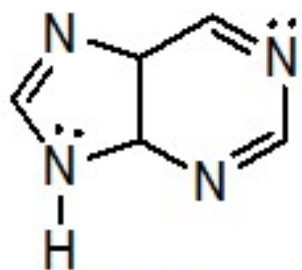
Furan



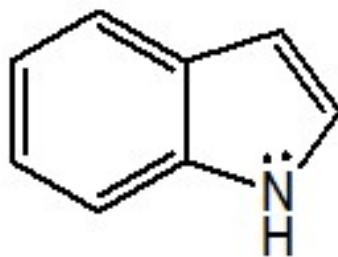
Thiophene

Fused Heterocyclic Aromatic Compounds

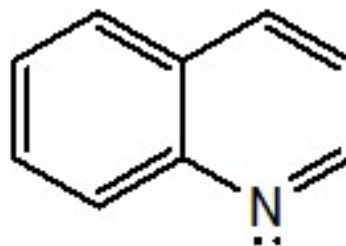
Having two rings sharing two atoms and the bond between them.



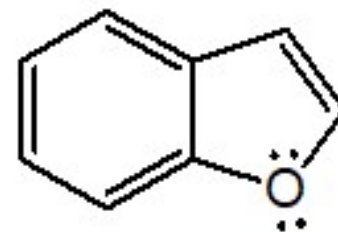
Purine



Indole

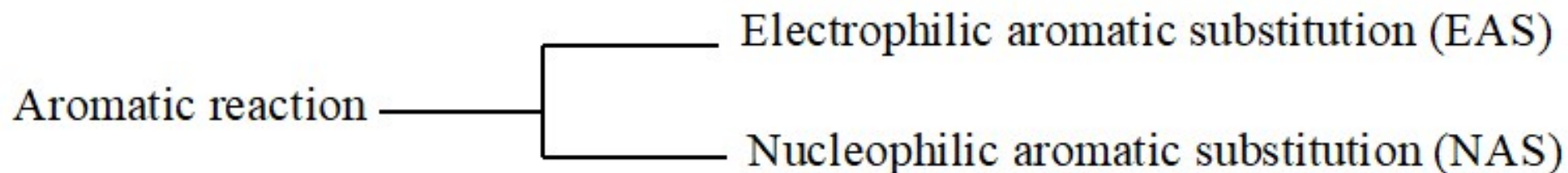


quinoline

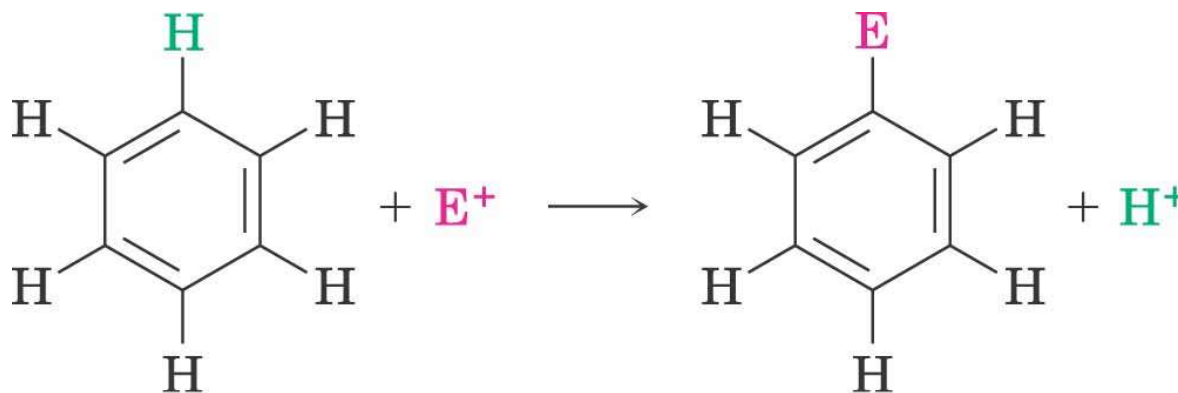


Benzofuran

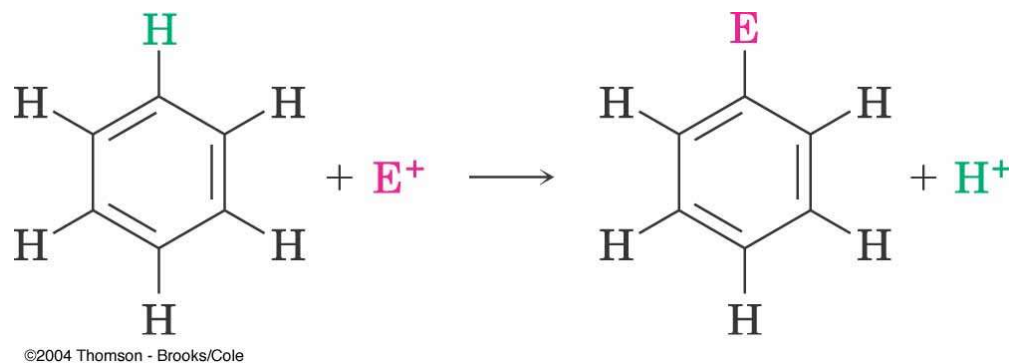
Aromatic Substitution Reactions



EAS- substitution of an electrophile (E^+) for a proton (H^+) on the aromatic ring.

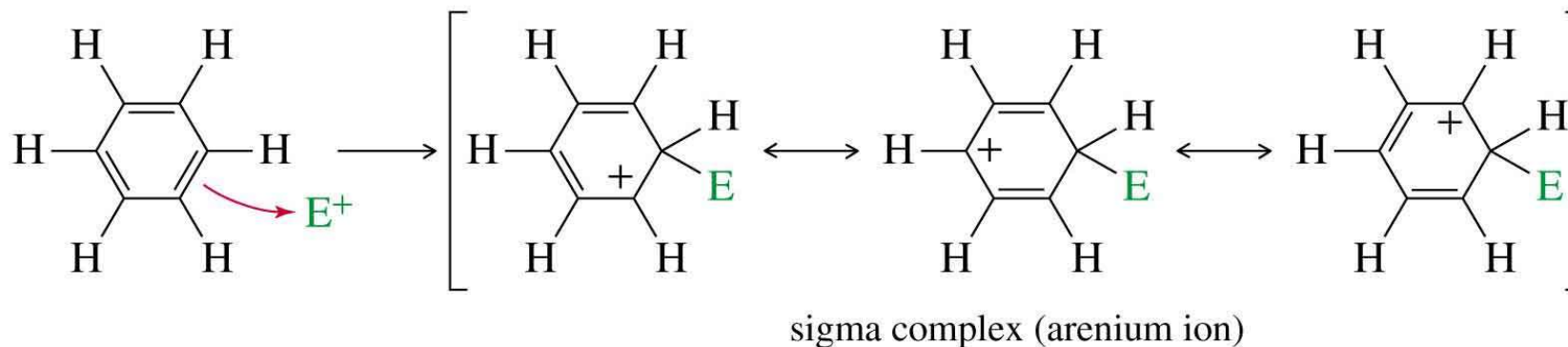


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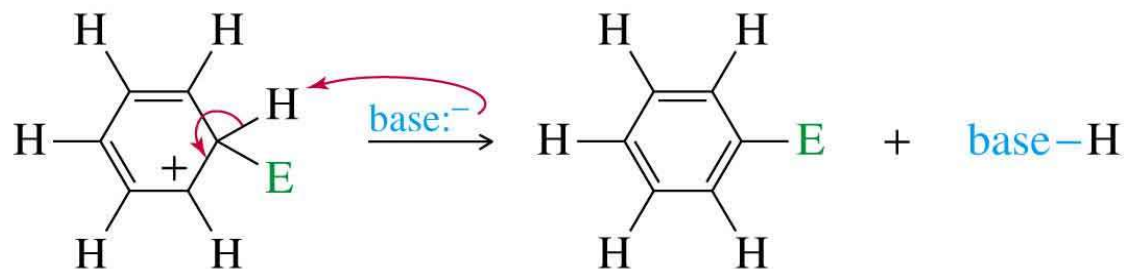


Mechanism

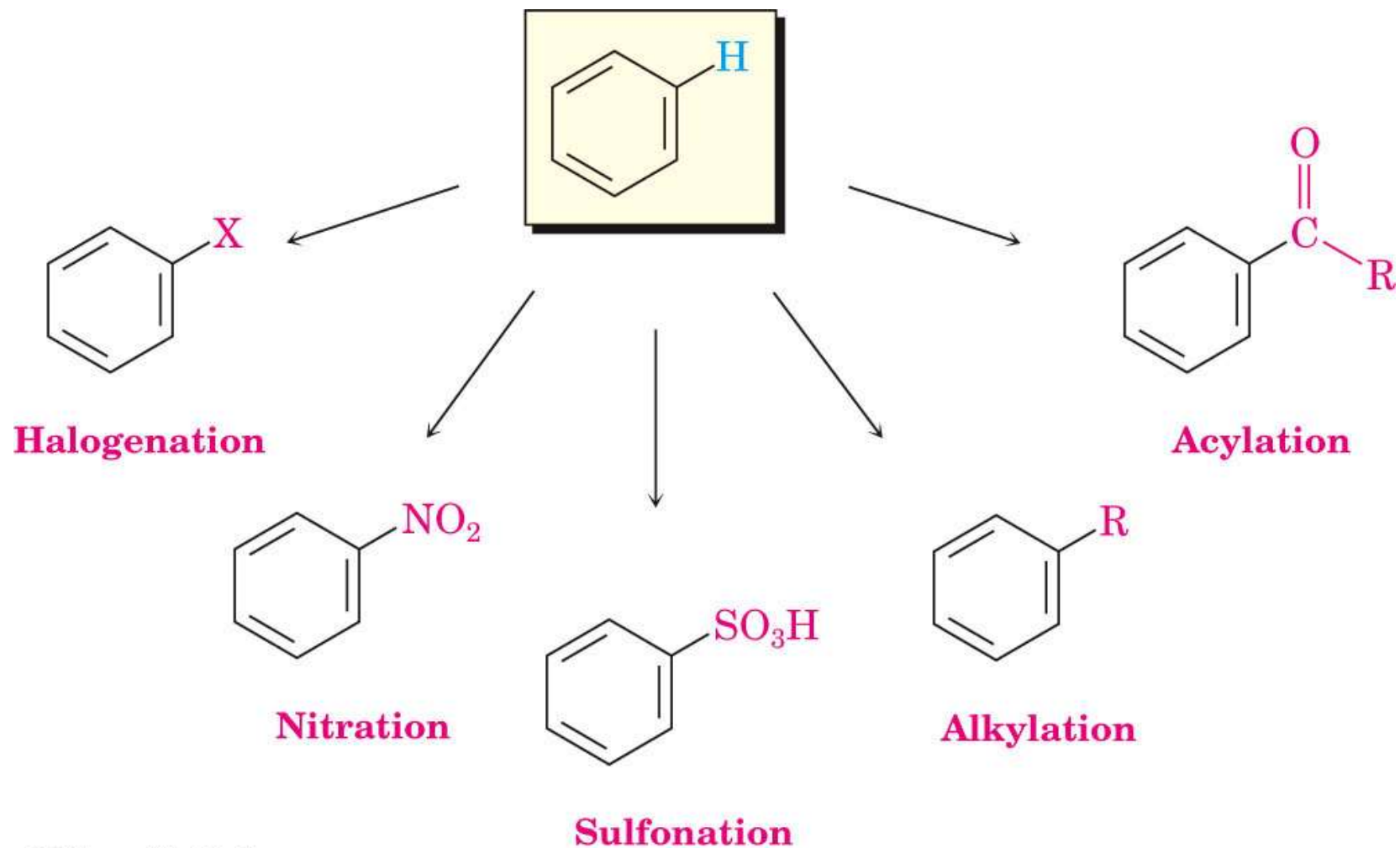
Step 1: Attack on the electrophile forms the sigma complex.



Step 2: Loss of a proton gives the substitution product.

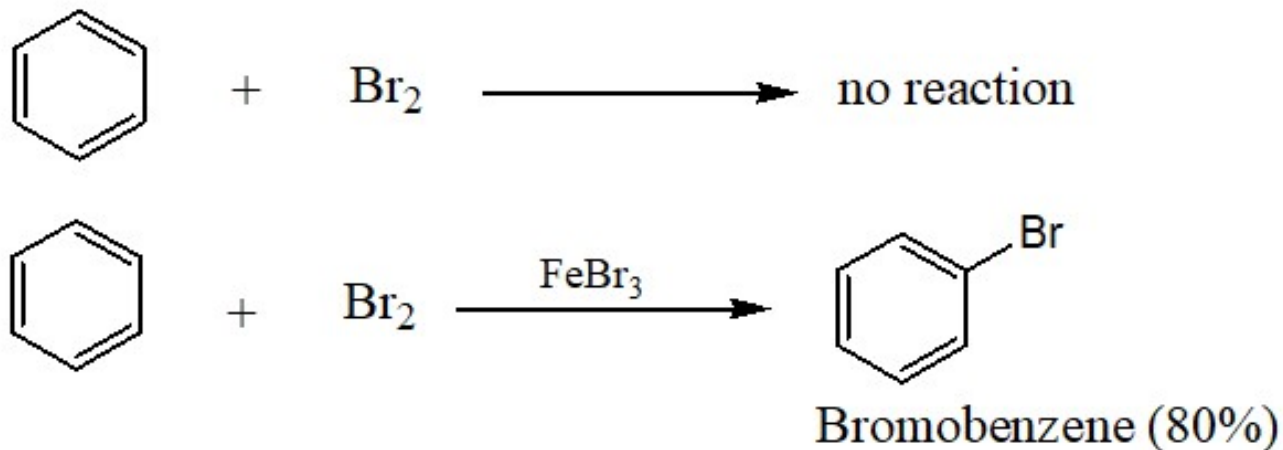


Some electrophilic aromatic substitution reaction (EAS)



Halogenation

Bromination



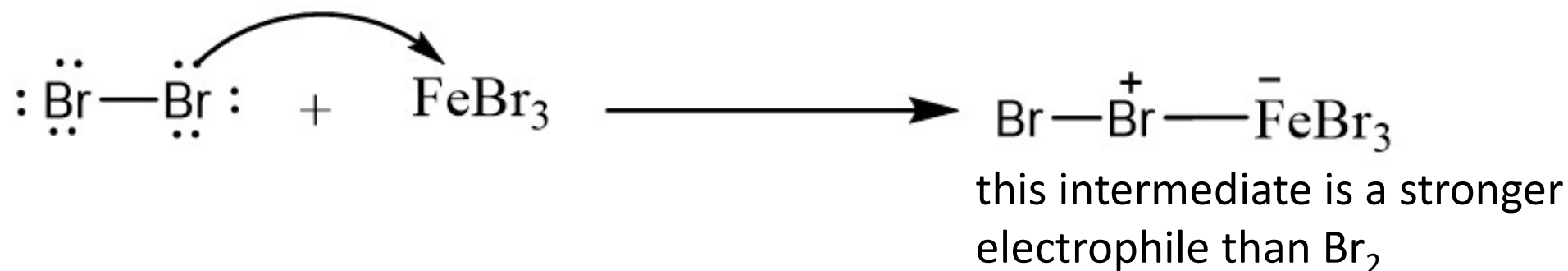
Br₂ is not sufficiently electrophilic to react with benzene and the formation of Br⁺ is difficult.

A strong Lewis acid such as FeBr₃ catalyzes the reaction.

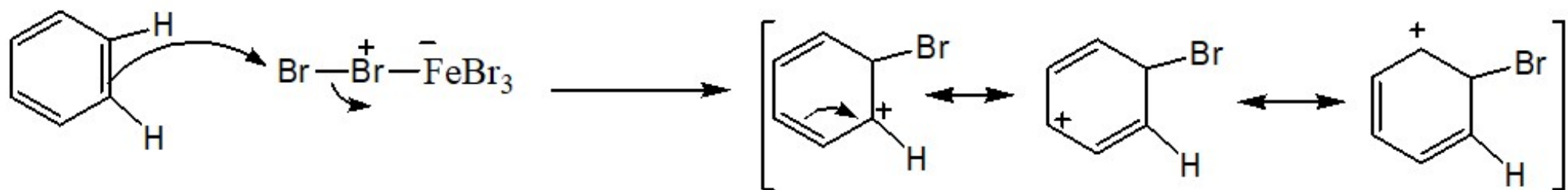
Catalysts in Halogenation: FeBr₃, FeCl₃, AlCl₃

Mechanism

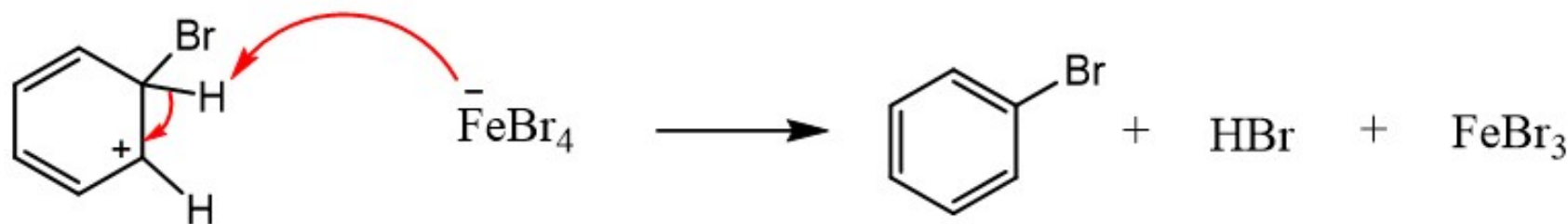
Step 1: formation of stronger electrophile.



Step 2: electrophilic attack and formation of sigma complex



Step 3: Loss of proton, regeneration of the aromatic ring

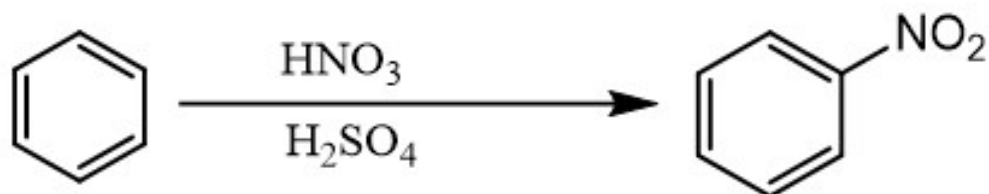


Chlorination

Use AlCl_3 as the Lewis acid catalyst.

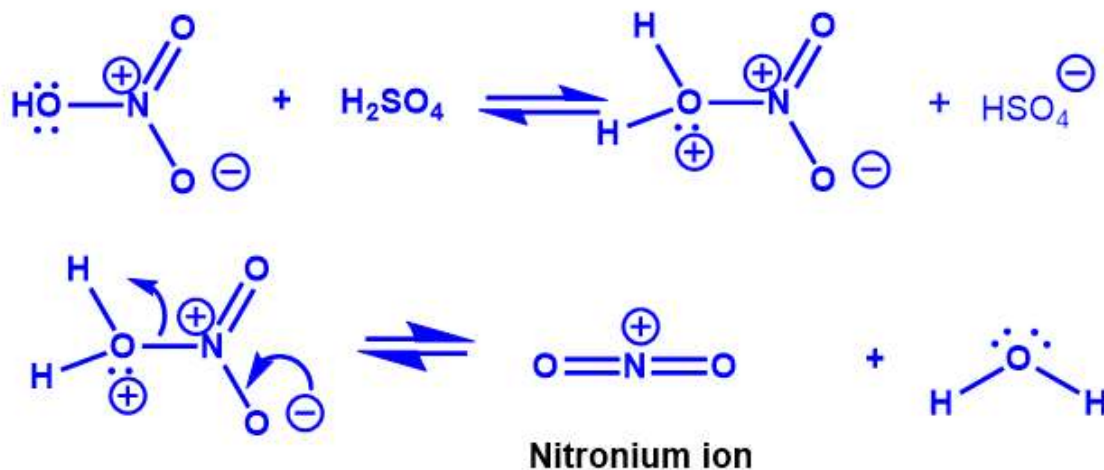


Nitration



The combination of nitric acid and sulfuric acid produces NO_2^+ (nitronium ion)

Step 1



Nitronium ion (NO_2^+) is the electrophile that reacts with the benzene ring.

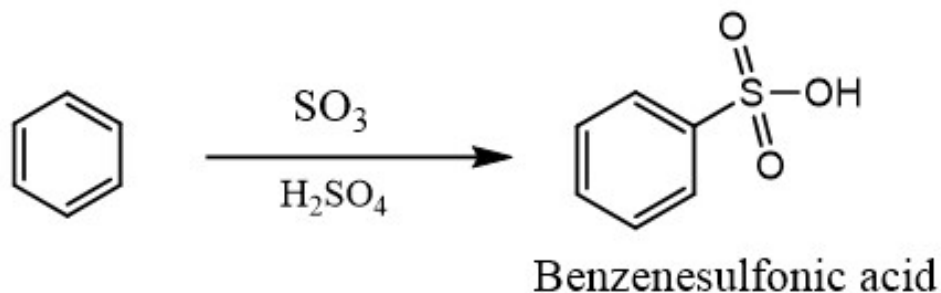
Step 2



Step 3



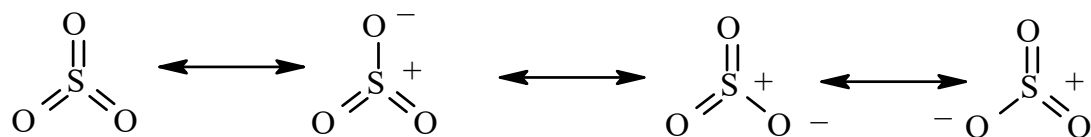
Sulfonation



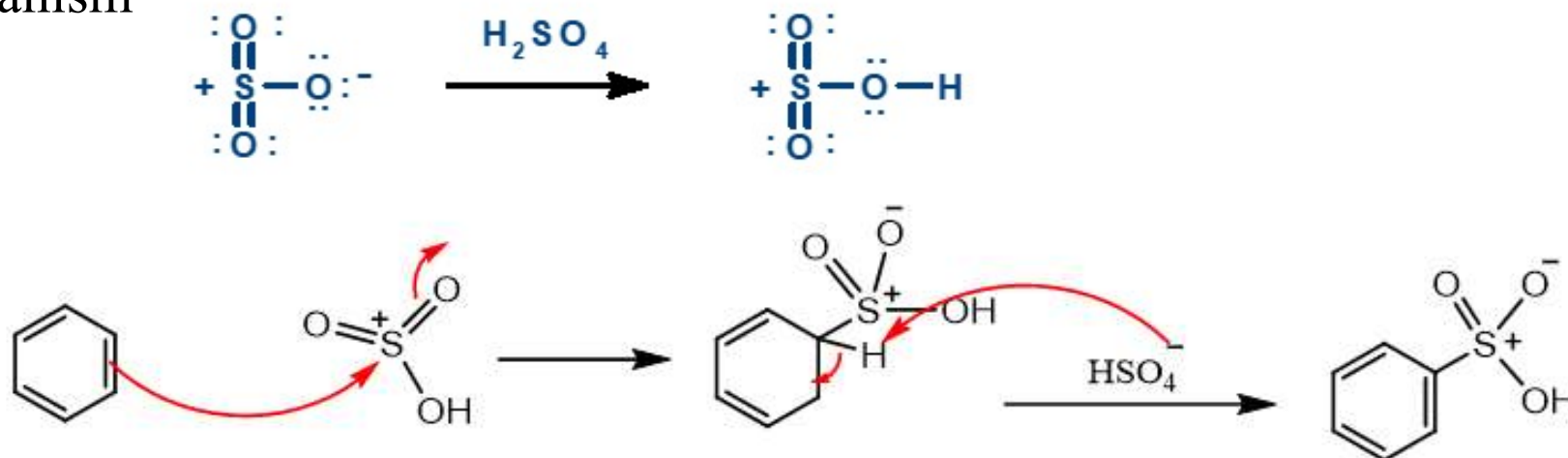
Reaction with a mixture of sulfuric acid and SO_3 (Fuming H_2SO_4)

The actual electrophile in this reaction is sulfur trioxide.

SO_3 is strong electrophile

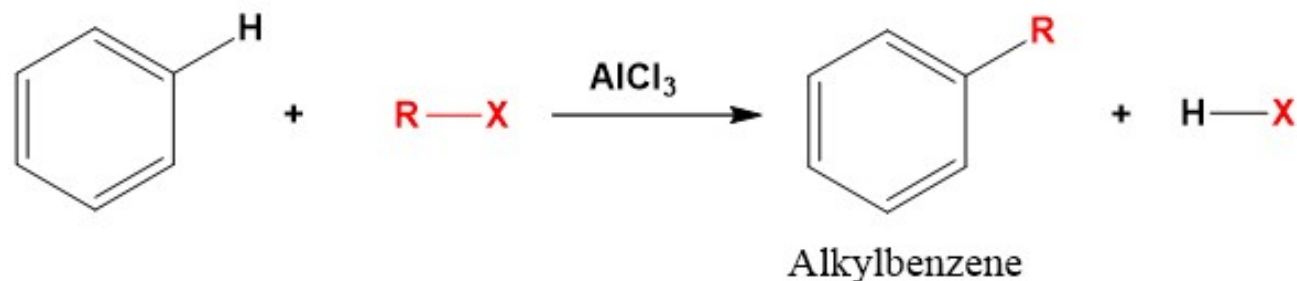


Mechanism

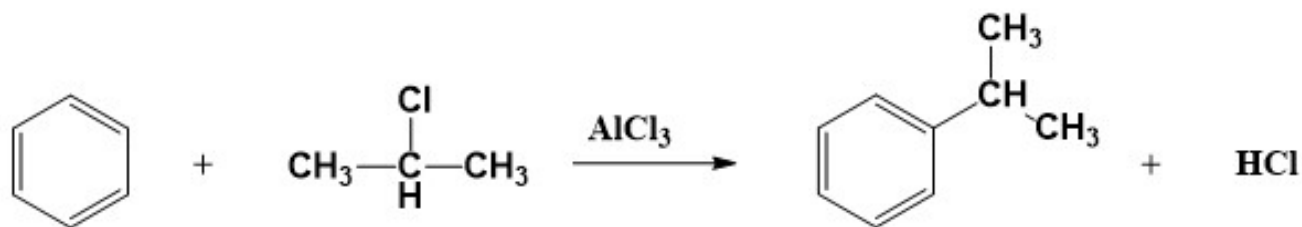
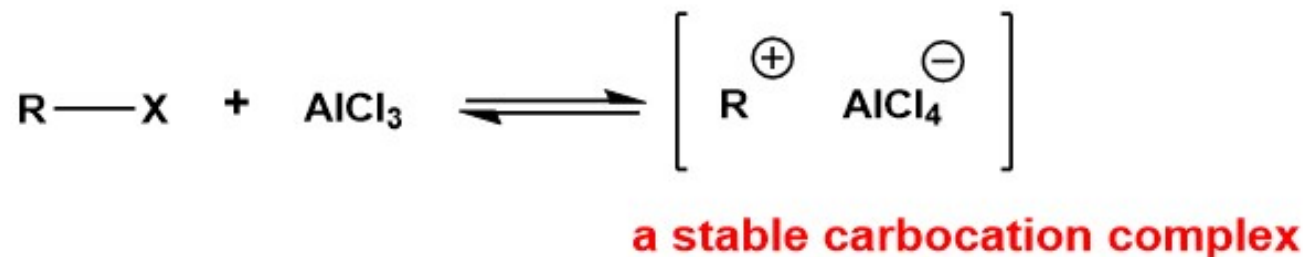


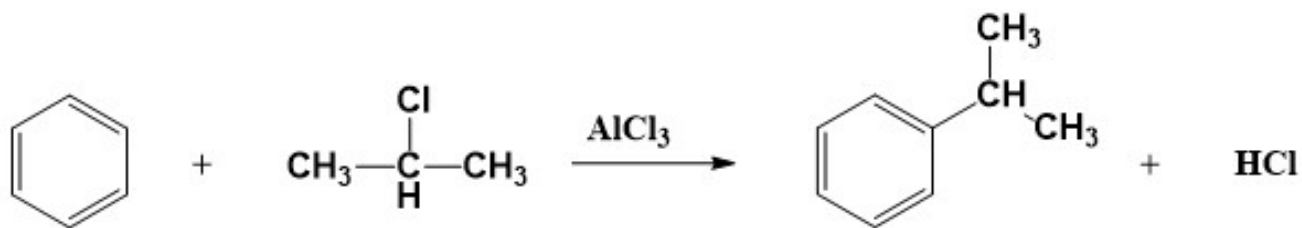
The Friedel-Crafts Alkylation

Synthesis of alkyl benzenes from alkyl halides and a Lewis acid, usually AlCl_3



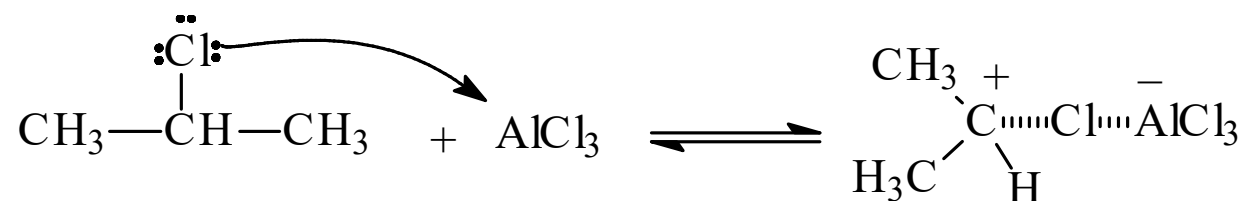
- The role of the AlCl_3 is to generate a stable carbocation complex.
- The carbocation is the electrophile in this process.



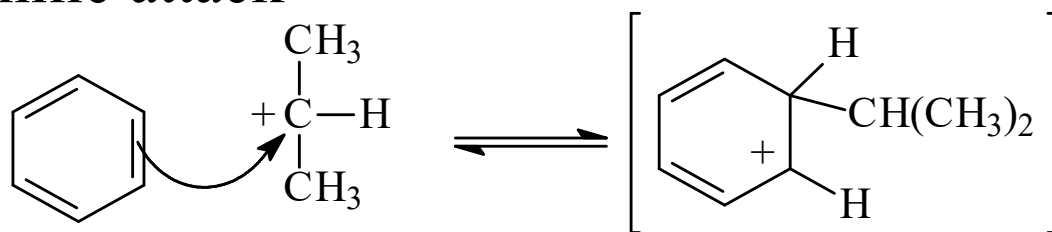


Mechanism

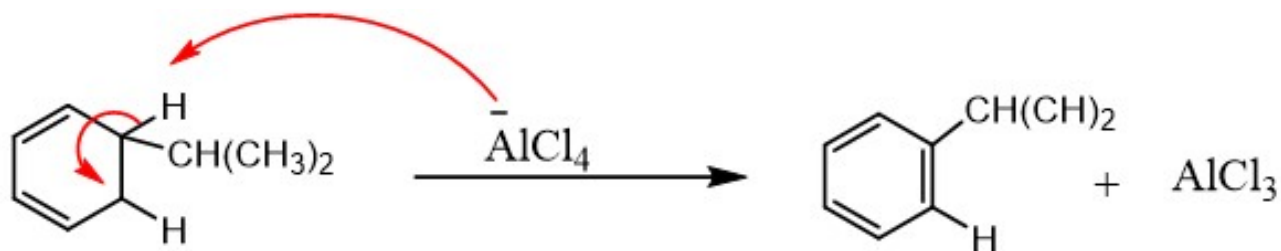
1. Formation of Carbocation



2. Electrophilic attack

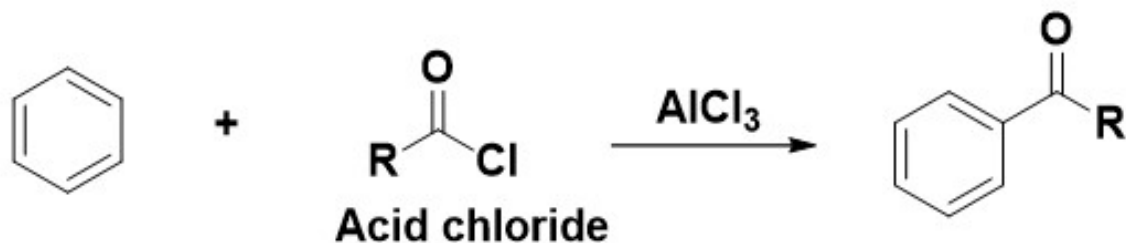


3. Loss of proton

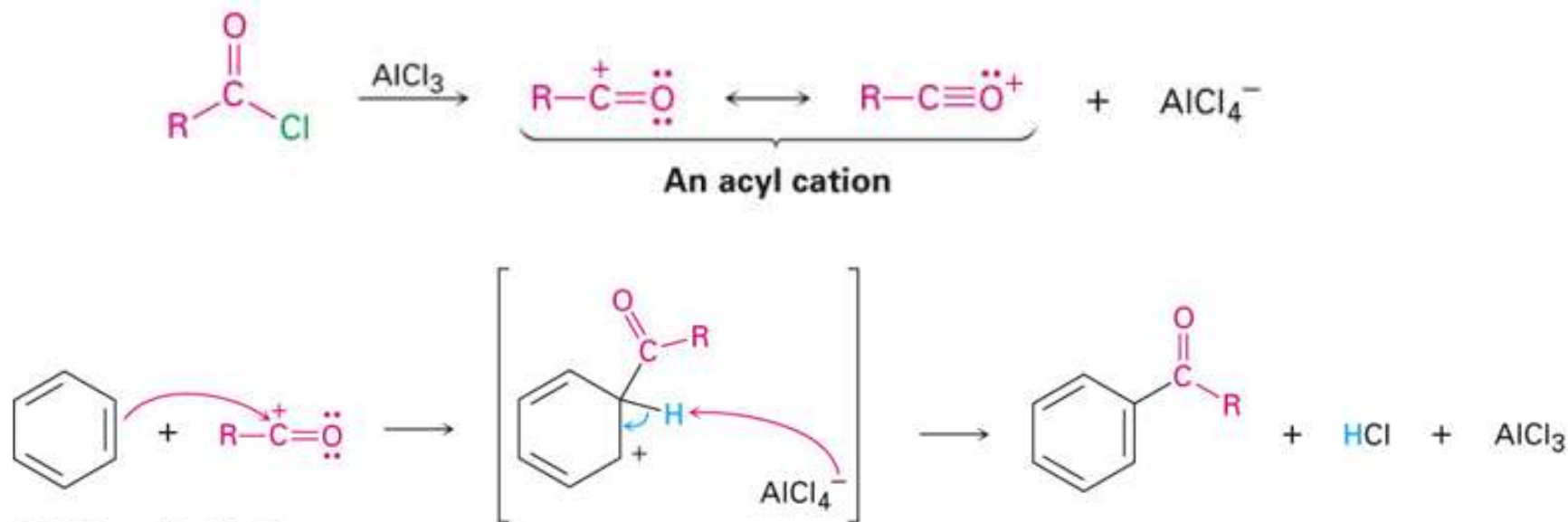


The Friedel-Crafts Acylation

Reaction of an acid chloride (RCOCl) and an aromatic ring in the presence of AlCl_3 introduces acyl group, —COR

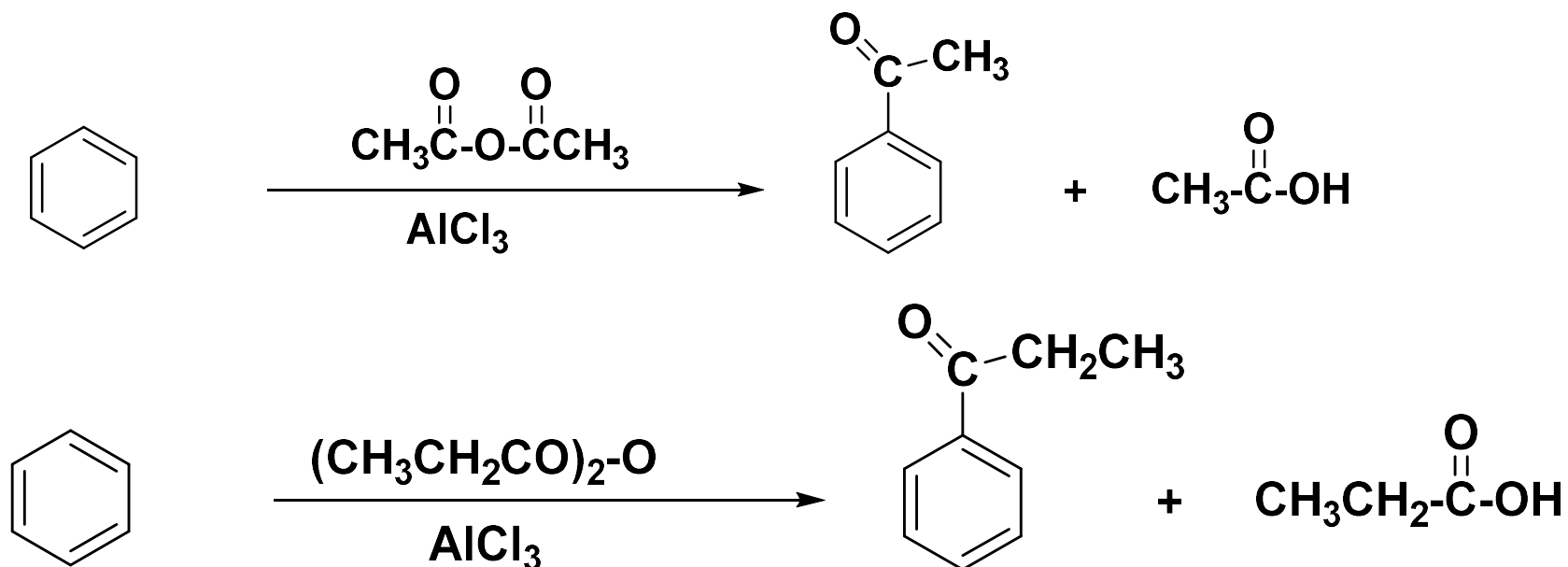


Mechanism



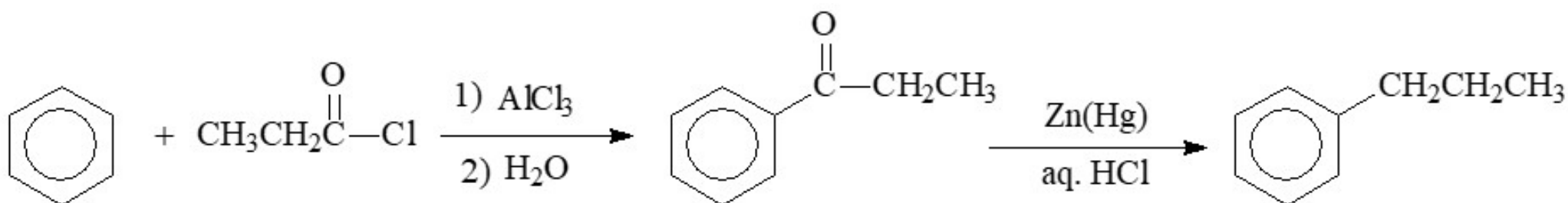


Acid anhydrides can also serve as a source of acyl cation, in the presence of AlCl_3 , acylate benzene.

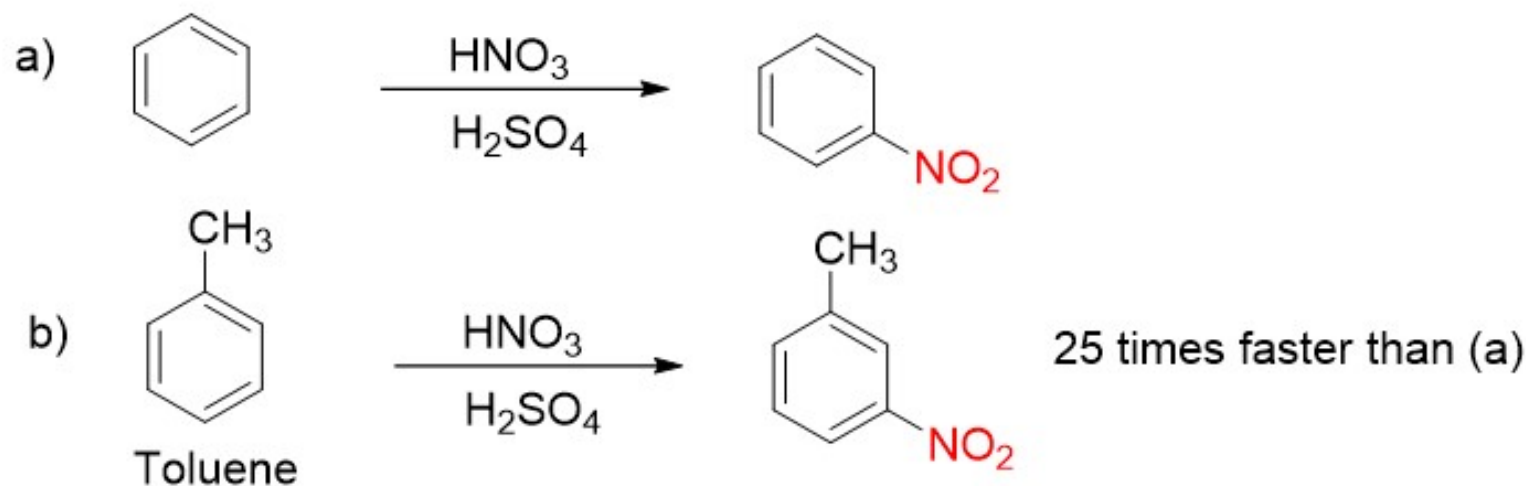


Clemmensen Reduction

Acylbenzenes can be converted to alkylbenzenes by treatment with aqueous HCl and amalgamated zinc.



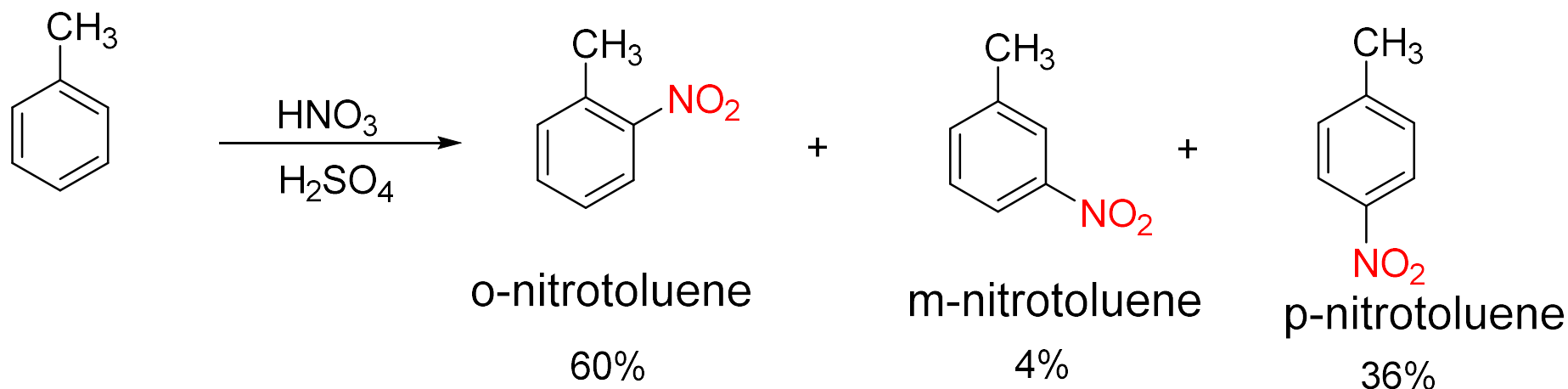
Directing Effects of Substituents



Toluene reacts 25 times faster than benzene under the same condition.

- Toluene is activated towards EAS.
- CH_3 is activating group.

Nitration of toluene gives a mixture of products.



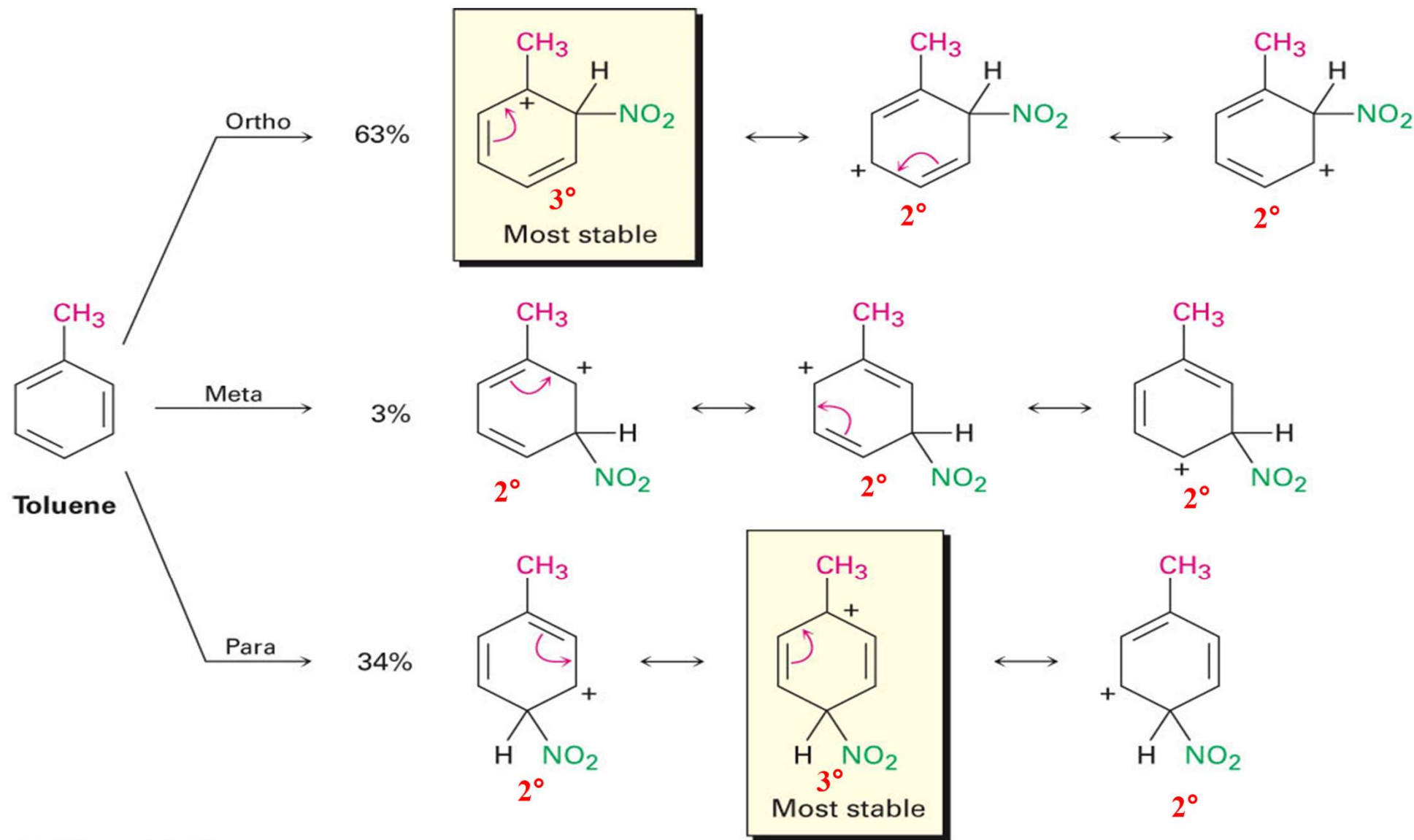
➤ Methyl group is ortho-para director

- Substituents can cause a compound to be more reactive or less reactive than benzene.
- Substituents affect the orientation of the reaction.
- Three groups: **ortho/para-directing** activators, **ortho/para-directing** deactivators, and **meta-directing** deactivators.

Ortho/Para-Directing Activators

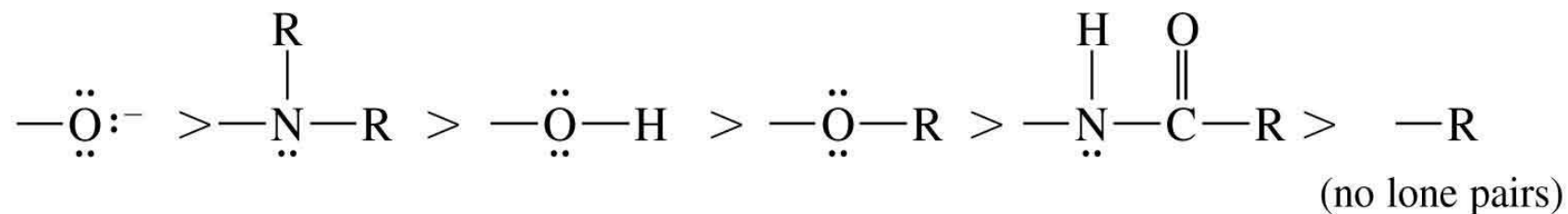
Alkyl Groups

- the ortho and para-attack produce more stable intermediate.

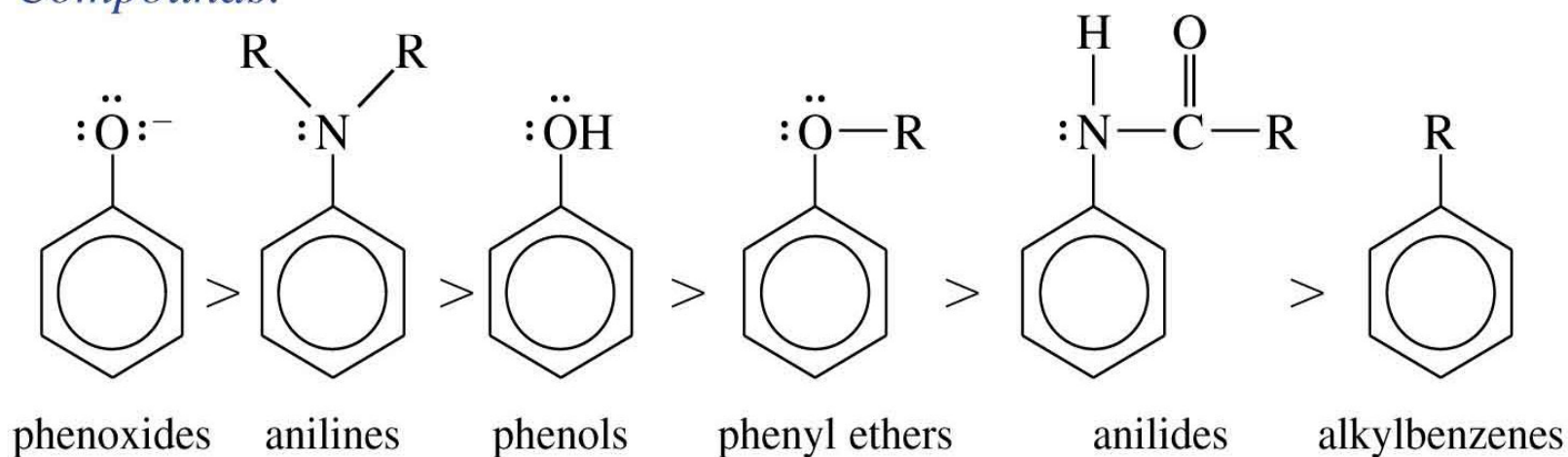


Ortho/Para-Directing Activators:

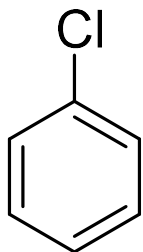
Groups:



Compounds:

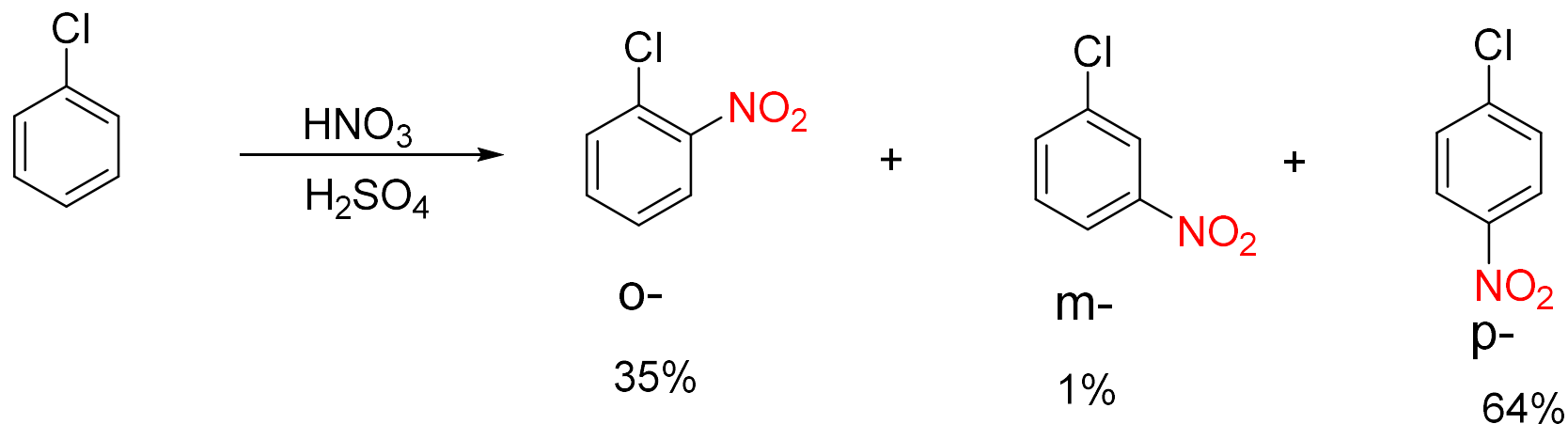


Ortho/Para-Directing Deactivators: Halogens

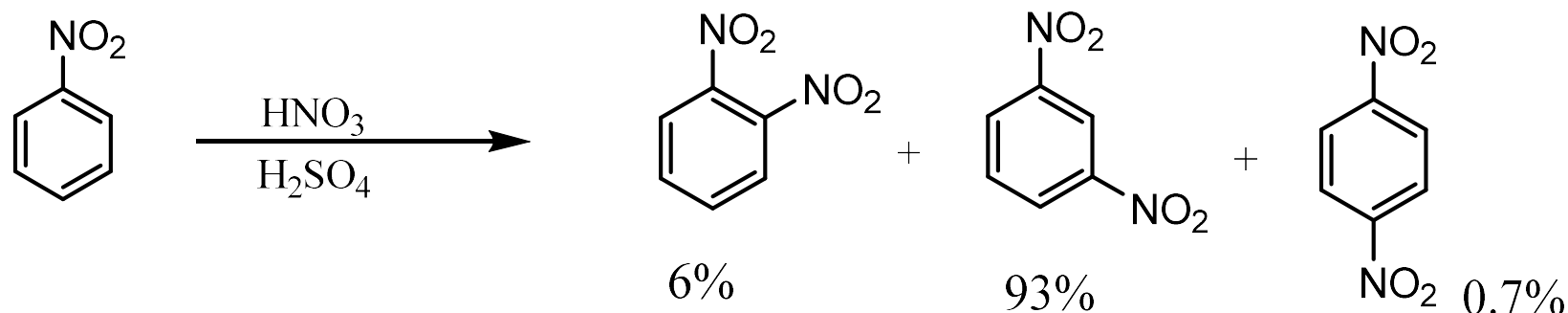


Ortho/Para-Directing Deactivators: Halogens

- Halogens are deactivating toward electrophilic substitution, but are ortho, para-directing.
- Since halogens are very electronegative, they withdraw electron density from the ring inductively along the sigma bond.



Meta-Directing Deactivators

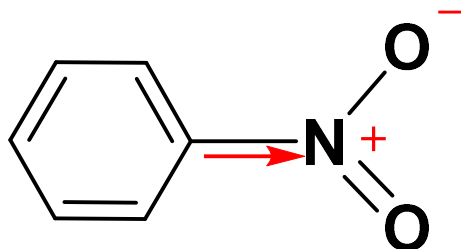


$\Rightarrow \text{NO}_2$ is meta directing

- The reaction was 100,000 times slower than for benzene.

$\Rightarrow \text{NO}_2$ is deactivating

Electron withdrawing groups withdraw electron density from the aromatic ring.



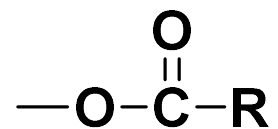
The aromatic ring is less electron rich than benzene

➤ Deactivated

Summary of Directing Effects

Ortho-para Directing

Activators

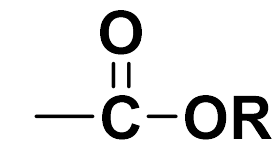
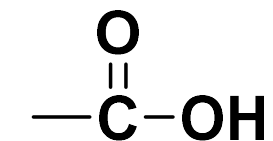
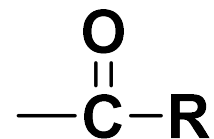


Deactivators



Meta-Directing

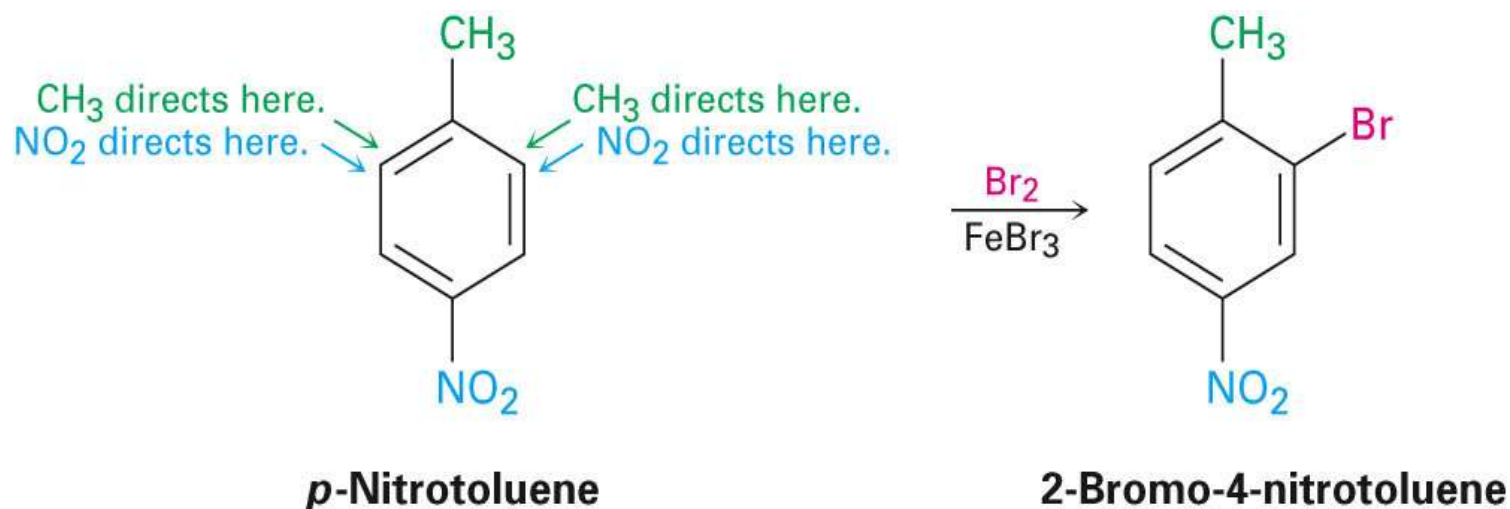
Deactivators



Effect of multiple substituents on EAS

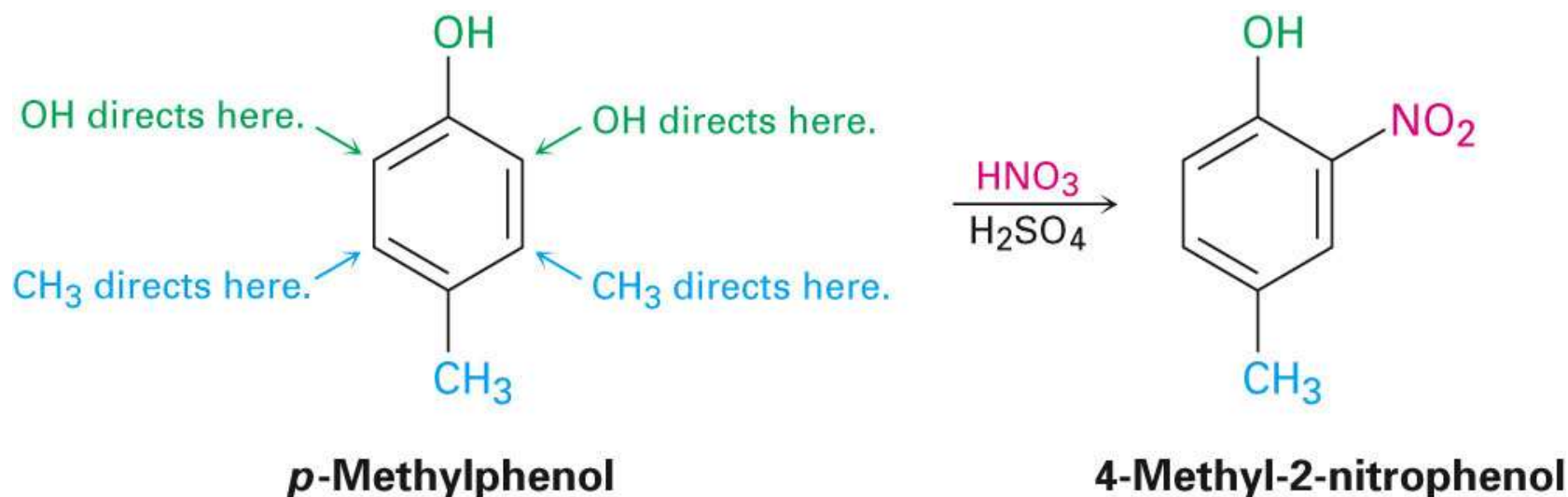
Activating groups are usually stronger directors than deactivating groups.

1. If the directing effects of the two groups are the same, the result is additive



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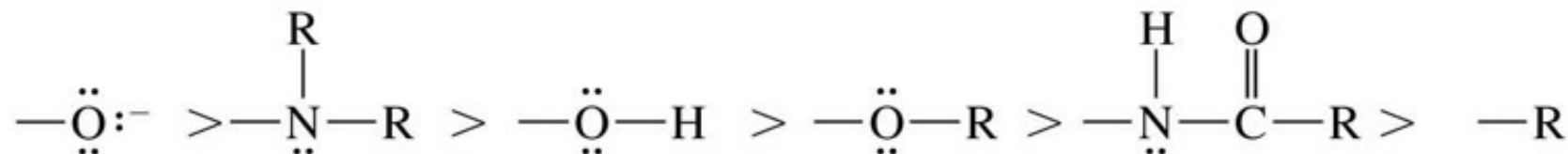
2. If the directing effects of two groups oppose each other, the more powerful activating group decides the principal outcome. Usually gives mixtures of products



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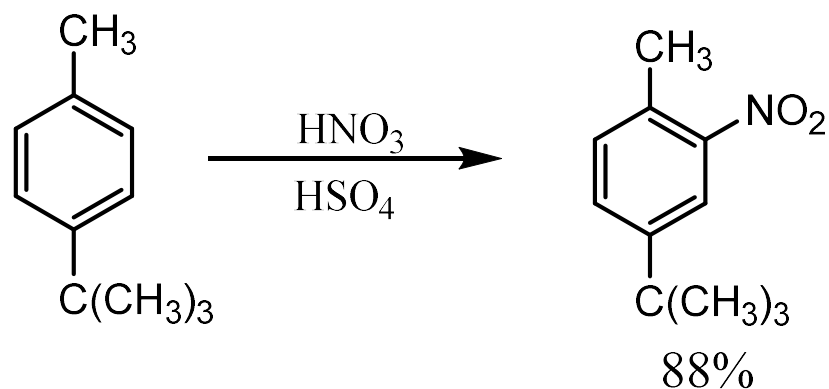
-OH is strongly o/p director than -CH₃

Powerful o/p directors: -OH, -OR, -NH₂, -OCOR



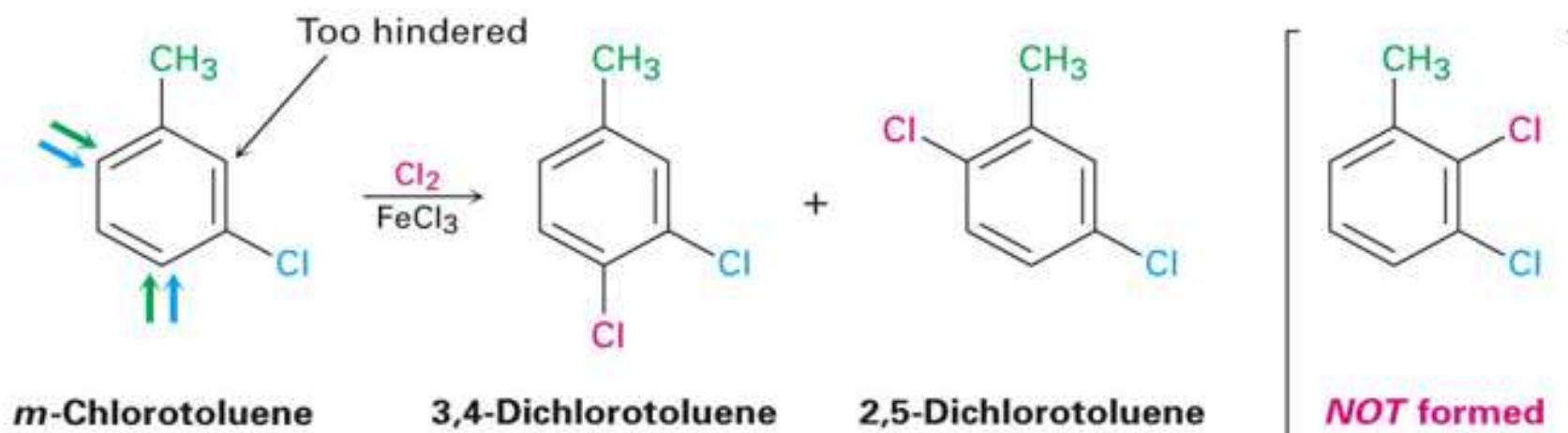
3. If the 2 groups direct to different sites; and activating effects are similar.

Substitution occurs ortho to the smaller group



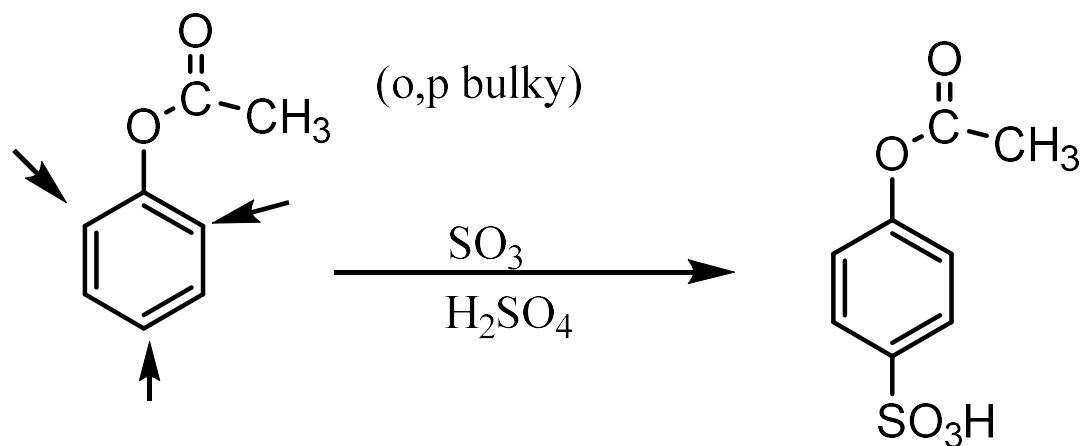
4. Substitution rarely occurs in between two substituents meta to each other.

Too hindered

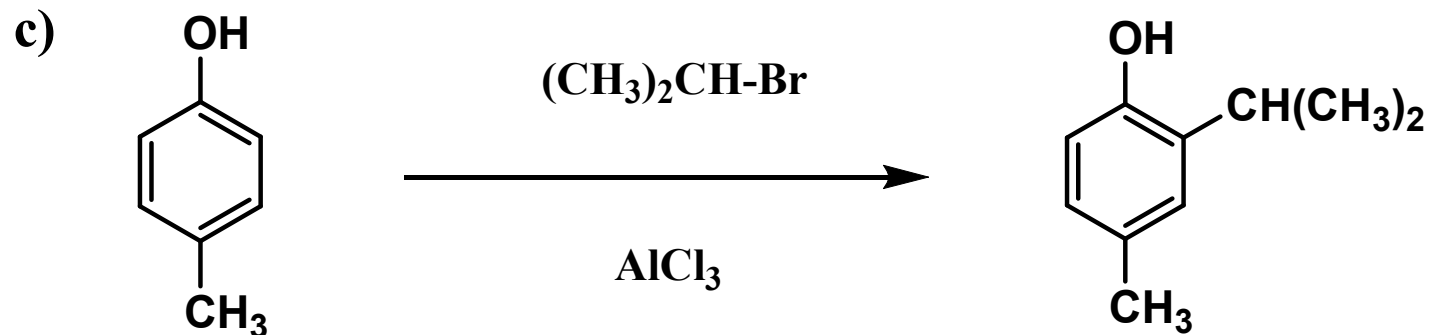
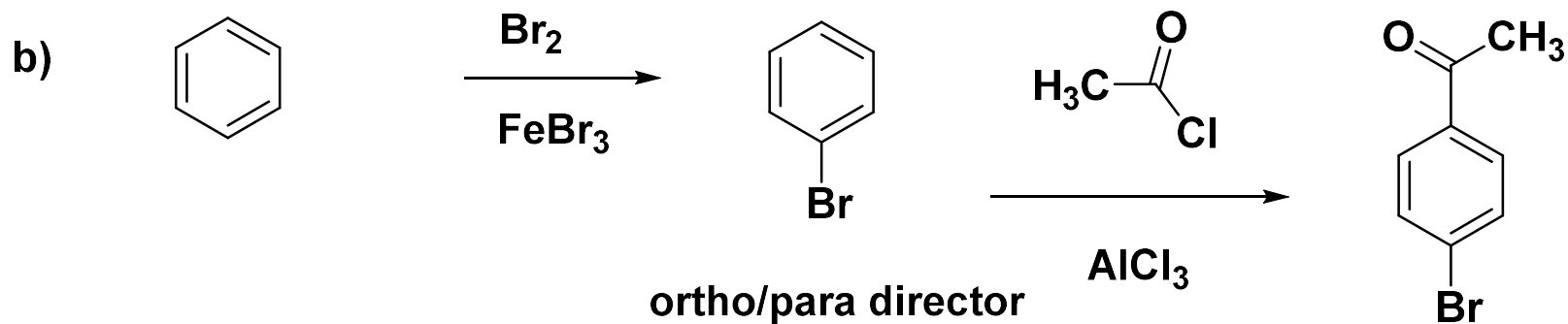
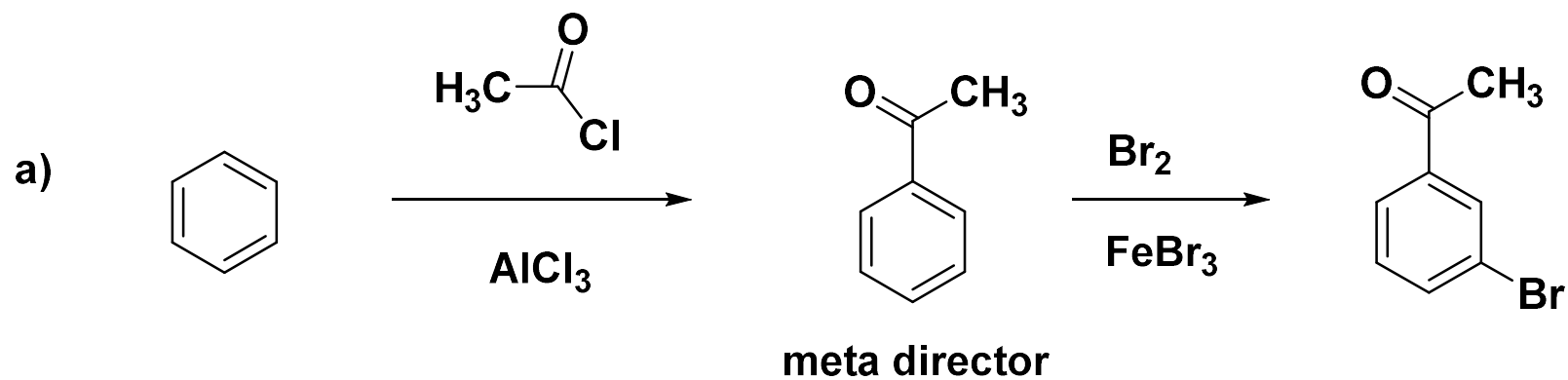




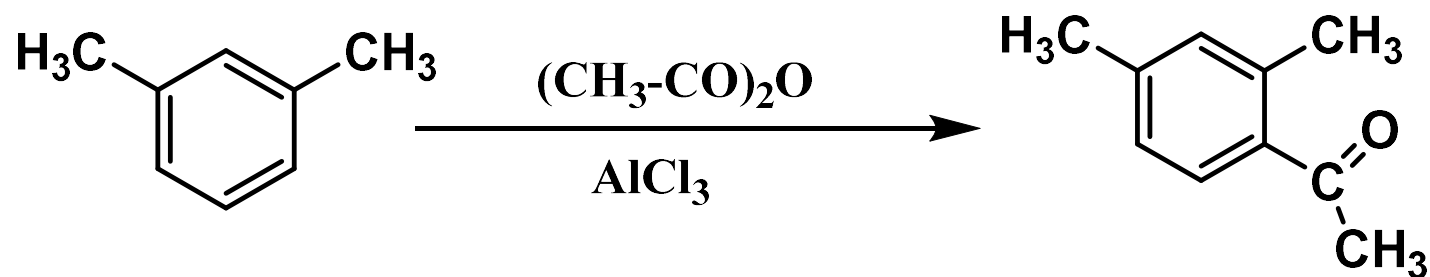
5. With a bulky o/p director and/or bulky electrophiles, para substitution predominates.



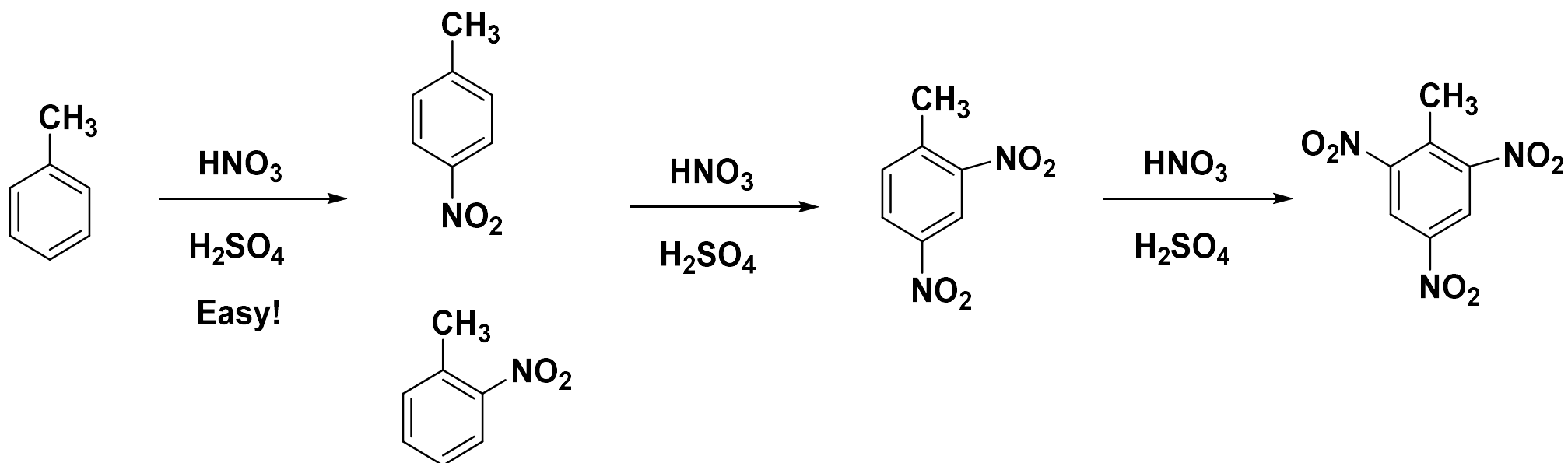
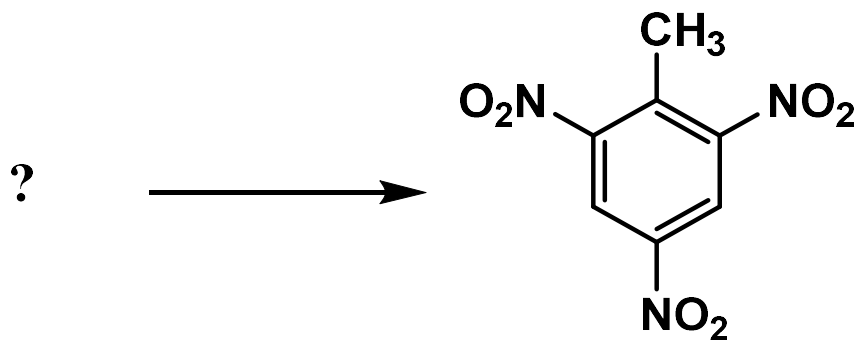
Examples of EAS



d)



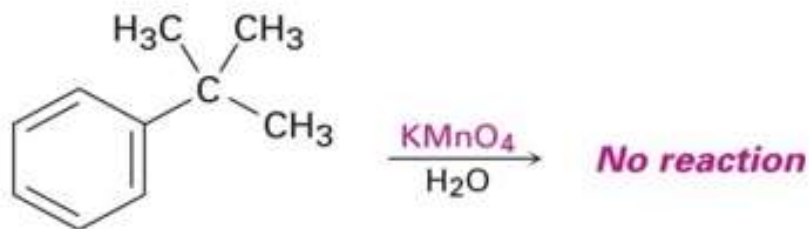
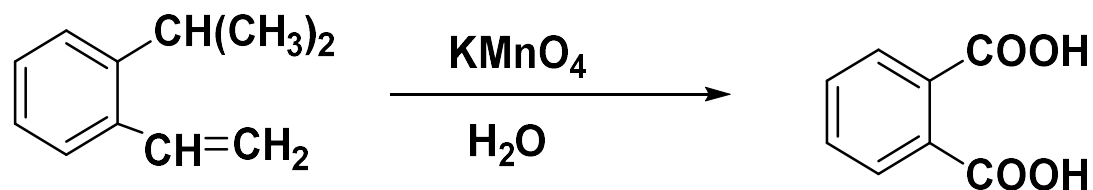
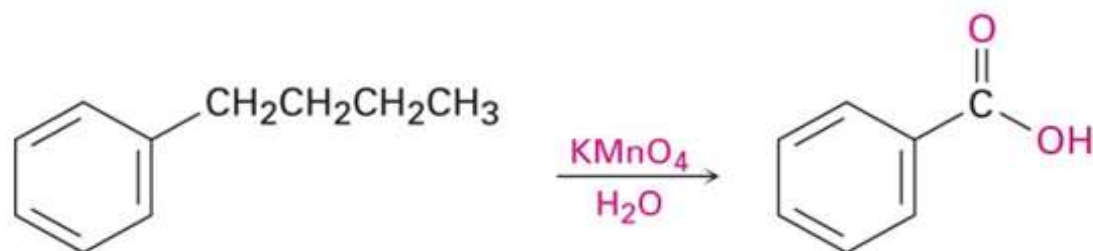
e)



Reactions of Aromatic Side Chains

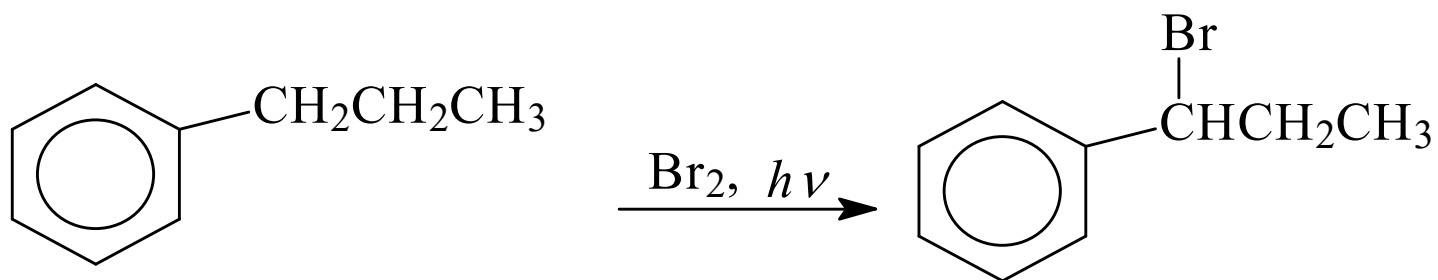
Oxidation of alkyl Side-Chain

- Alkyl side chains can be oxidized to $\text{—CO}_2\text{H}$ by strong reagents such as hot KMnO_4 or $\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$ if they have a C-H next to the ring



Side-Chain Halogenation

- Benzylic position is the most reactive.
- Chlorination is not as selective as bromination, results in mixtures.
- Br_2 reacts only at the benzylic position.



Hydrolysis of Sulfonic Acids

- Desulfonation can take place with simple hydrolysis

